

Study Guide

COMPOSITION OF EXAM

Lecture topic	Points	# of Qs	% Total points	% Total questions
Research Methods	9	4	6.77	7.55
Genetics & Evolution	10	4	7.52	7.55
Motor System	18	11	13.53	20.75
Visual	35	12	26.32	22.64
Auditory	20	5	15.04	9.43
Somatosensory	16	5	12.03	9.43
Gustatory	6	2	4.51	3.77
Olfactory	5	4	3.76	7.55
General Sensory	6	2	4.51	3.77
Wake/Sleep	8	4	6.02	7.55
TOTAL	133	53	100.00	100.00

Lecture 7 Research Methods

- Optogenetics and chemogenetics are 2 techniques used to manipulate neuronal activity

Record Brain Activity during Behavior	
Record from electrodes in brain	Invasive; used with laboratory animals, rarely with humans
Electroencephalograph (EEG)	Records from scalp; measures changes by milliseconds, but with low resolution of location of the signal
Evoked potentials	Similar to EEG but in response to stimuli
Magnetoencephalograph (MEG)	Similar to EEG but measures magnetic fields
Positron emission tomography (PET)	Measures changes over both time and location but requires exposing brain to radiation
Functional magnetic resonance imaging (fMRI)	Measures changes over about 1 second, identifies location within 1 to 2 mm
Correlate Brain Anatomy with Behavior	
Computerized axial tomography (CAT)	Maps brain areas, using X-rays
Magnetic resonance imaging (MRI)	Maps brain areas in detail, using magnetic fields

Lecture 8 Genetics & Evolution

- Chromosomes: DNA is wrapped around **histones** and coiled.
- Genes: Sections of DNA that code for making proteins
- **Alleles** - two forms of the same gene

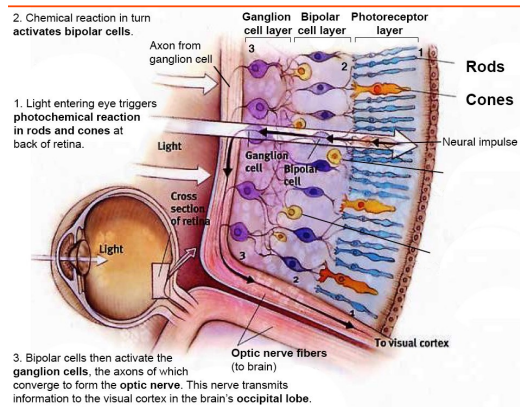
- Epigenetics - the study of how experiences and behaviors influence the expression of genes
- Know how the axons grew back when Sperry cut the newt's optic nerve

Types of sensory receptors

- Mechanoreceptors
 - o Somatosensory
 - Meissner's corpuscle
 - Pacinian corpuscle
 - Merkel's disks
 - Ruffini organ
 - o Auditory
 - Hair cells (Organ of Corti)
 - o Vestibular
 - Hair cells (ampulla & otolith organ)
- Chemoreceptors
 - o Gustatory
 - Taste receptors (5 types)
 - 3 are GPCRs
 - 2 are ionotropic
 - o Olfactory
 - Olfactory receptors
 - GPCRs
- Photoreceptors
 - o Rods and cones of the retina
 - GPCRs

Lecture 9 Vision

- Stimulus
 - Know the properties of light and its related perceptions
- Anatomy of the Eye
 - Know the cranial nerves associated with the eye (related to sensory and motor information)
- Retina
 - Know the organization of the different cell types in the back of the retina



- Know the properties of cones and rods

Rods (scotopic vision)	Cones (photopic vision)
night vision (very sensitive)	day vision, other bright light
more adapted to darkness/dim light but takes time (slow recovery)	high visual acuity, color (fast recovery time)
more (120 million)	fewer (6 million)
multiple rods converge their information onto a single bipolar cell	cones have a 1:1 convergence onto bipolar cells (fovea)
located in the periphery	located in and near the fovea

- Know the function of bipolar and ganglion cells
- Understand the meaning of receptive fields and how it changes between cell types
 - Receptive fields get larger as we progress from the retina to cortex
- Be able to define lateral inhibition and which cell type is responsible for it
- Optic disk
 - Know what it is and its associated properties
- Optic chiasm
 - Know where the information goes when light strikes the right/left visual field and/or right/left eye (IN-CLASS ACTIVITY)
 - Know what is impaired when certain locations of the visual pathway is damaged
- Lateral geniculate nucleus
 - Know the two cell types and what information they carry

Parvocellular neurons

- info from midget retinal ganglion neurons (located in or near fovea)
- visual details
- color (red, green) “what”
- only to LGN
- small receptive fields

Magnocellular neurons

- info from parasol retinal ganglion neurons (located throughout retina) where
- movement
- overall patterns (very sensitive to contrast between light dark)
- to LGN, other thalamic areas
- large receptive fields

- Neuromuscular junction (NMJ)
 - Know the neurotransmitter used at the NMJ
- Proprioception
 - Know the definition of a proprioceptor
 - Be able to name two proprioceptors and explain their function using an example
- Pyramidal system
 - Know the function of the primary motor cortex and its association motor areas
 - What brain areas are part of the association motor cortices
 - Know the two types of corticospinal tracts we discussed in class, what body parts the control
 - Lateral -contralateral limbs
 - Medial – medial body parts (e.g., trunk, neck, shoulder)
- Extrapyramidal system
 - Name the two brain regions involved in the extrapyramidal system
 - Describe their function
- Parkinson's disease
 - What are characterizations of Parkinson's disease
 - What brain regions are involved
 - What neurotransmitter is involved
 - How is the function of the basal ganglia altered

What are treatment options for Parkinson's disease and what do they do

Lecture 11 Auditory & Somatosensory Systems

Auditory System

- Properties of stimulus (sound)
 - Know the properties of sound and its related perceptions
- Structure of the ear
 - Know the various structures that make the outer, middle, and inner ear and their associated functions
- Inner ear
 - Know the structure, properties, and function of the Cochlea
 - Know the structure, properties, and function of the Organ of corti
 - Know what hair cells are, where they are located, and how they convert vibrations into a nerve impulse
- Auditory ear pathways
 - Know the events that happen from sound waves entering the ear to being converted into nerve impulses by the inner ear
- Auditory cortex
 - Know the properties of the auditory cortex
- Auditory brain pathway
 - Know the auditory pathways from the cochlea to the primary auditory cortex
- Deafness
 - Know and describe the different types of deafness we talked in class about, their cause, and potential treatment options

Vestibular System

- Structure and function
 - Know the structure and function of the vestibular system
- Physiology
 - Know what the vestibular receptors are, where they are located, and how they are activated
- Be able to discuss how motion sickness can be explained

Somatosensory system

- Stimulus
 - Know the various stimuli that the somatosensory system responds to
- Somatosensory receptors
 - Know the various somatosensory receptors and their corresponding function
 - How are these receptors activated?
- Somatosensory representation
 - Know how somatosensory information is represented in the somatosensory cortex
- Somatosensory pathways

Know the pathway from touch and how it differs from the pathway of pain

Lecture 12 Taste and Smell Systems

Taste

- Tastes
 - What are the primary tastes for gustation
- Taste buds & receptors
 - What are different types of papillas and where are they located
 - What are taste receptors and where are they located
- Transduction
 - Know how the different taste receptors are activated
- Gustatory pathways
 - Name the three cranial nerves that innervate the tongue and be able to locate them
 - Know how gustatory information travels to the brain
 - Know which brain structure is the primary gustatory cortex
- Gustation and Olfaction
 - How do both system work together
 - What brain regions is innervated by taste and smell axons

Olfaction

- Smell
 - What are the primary smells for olfaction
- Anatomy of nose
 - Where is the olfactory system located
- Olfactory receptors
 - What are olfactory receptors and where are they located
- Transduction

- Know how olfactory receptors are activated and how they send information to the brain
 - Olfactory pathways
- Know how olfactory information travels to the brain

Lecture 13 Wakefulness & Sleep

- Circadian rhythm
 - Define circadian rhythm, give examples
 - How do they differ from other endogenous rhythms?
 - What experiments are conducted to show a circadian rhythm exists?
- Biological clock
 - What brain region is the main driver of the circadian rhythm/our biological clock?
 - What experiments prove that that brain region is indeed the main driver?
 - What are the underlying mechanisms (know the different proteins involved and how they are up- or down-regulated at night/daytime, etc.)
 - How can light entrain the biological clock/the brain region?
 - What hormone increases sleepiness and which region releases this hormone?
- Stages of sleep
 - Know the different stages and how they differ from each other when assessed by EEG
- Pattern of sleep stages
 - How do people “normally” progress through the different stages of sleep?
- Brain mechanisms
 - What brain regions and neural systems underly the different sleep stages?
- Sleep disorders
 - What are the two different sleep disorders we talked about in class?
 - Name what causes these disorders and what treatments are available.
- Sleep theories
 - What are different theories that explain the function of sleep?
 - Know what is meant by sleep is necessary to restore mind.
 - What are evidence that shows sleep aids memory?